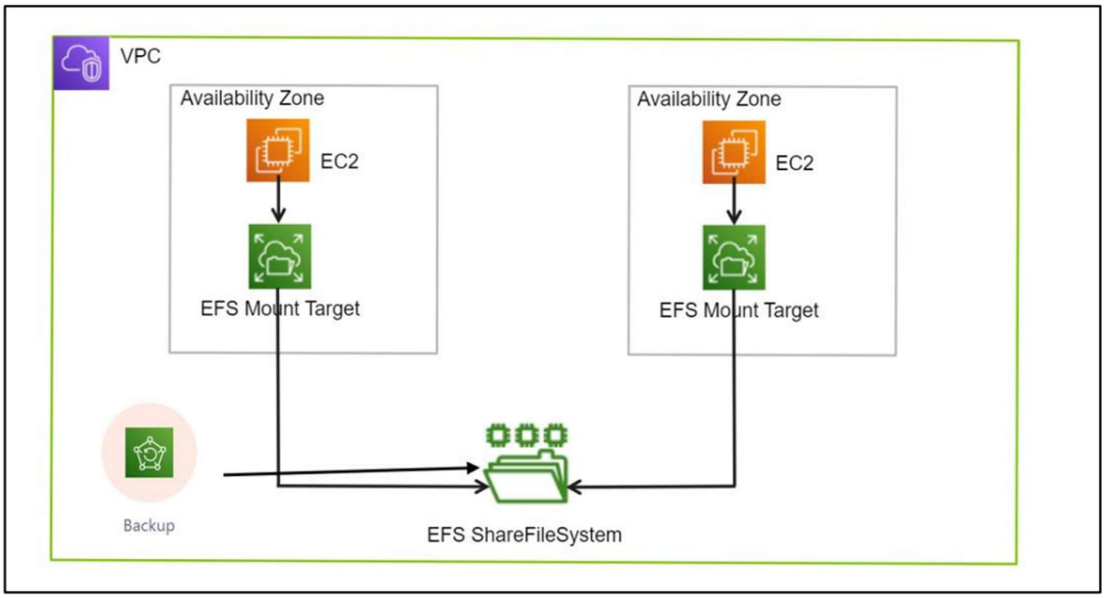
Case Study 3 - Backup Strategy to ensure mission critical data is being backed up regularly

Problem Statement:

The media hosting organization want to setup the backup plan for application servers, EFS. You should have a strategy to backup these data sources periodically using AWS Backup, and finally, automate the testing of the restore process.

a) Create a Backup Strategy to ensure mission-critical data is being backed up regularly
b) Test restoring from backups to ensure there are no data recovery issues
c) Learn how to automate this process.

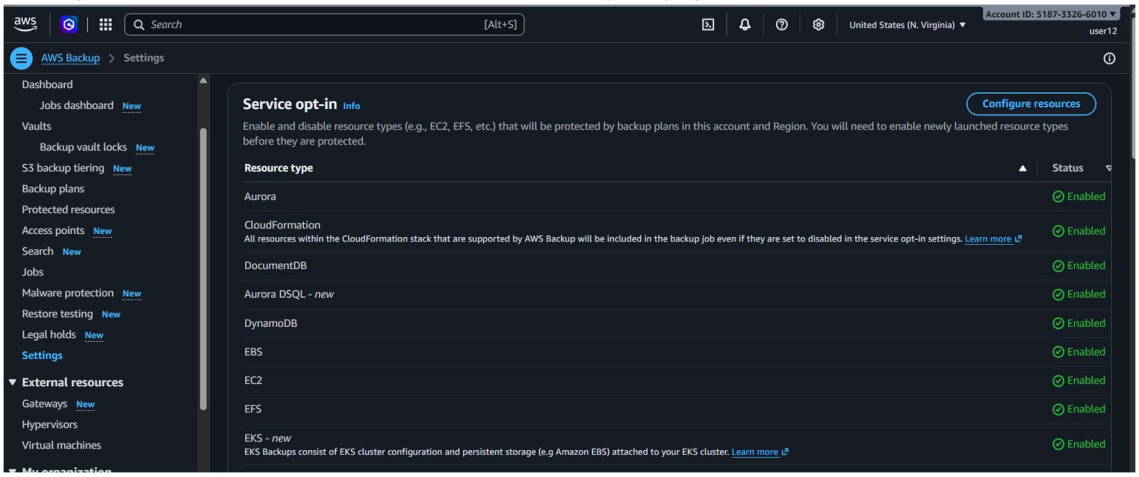
Architecture of AWS Backup for EFS is as follows:



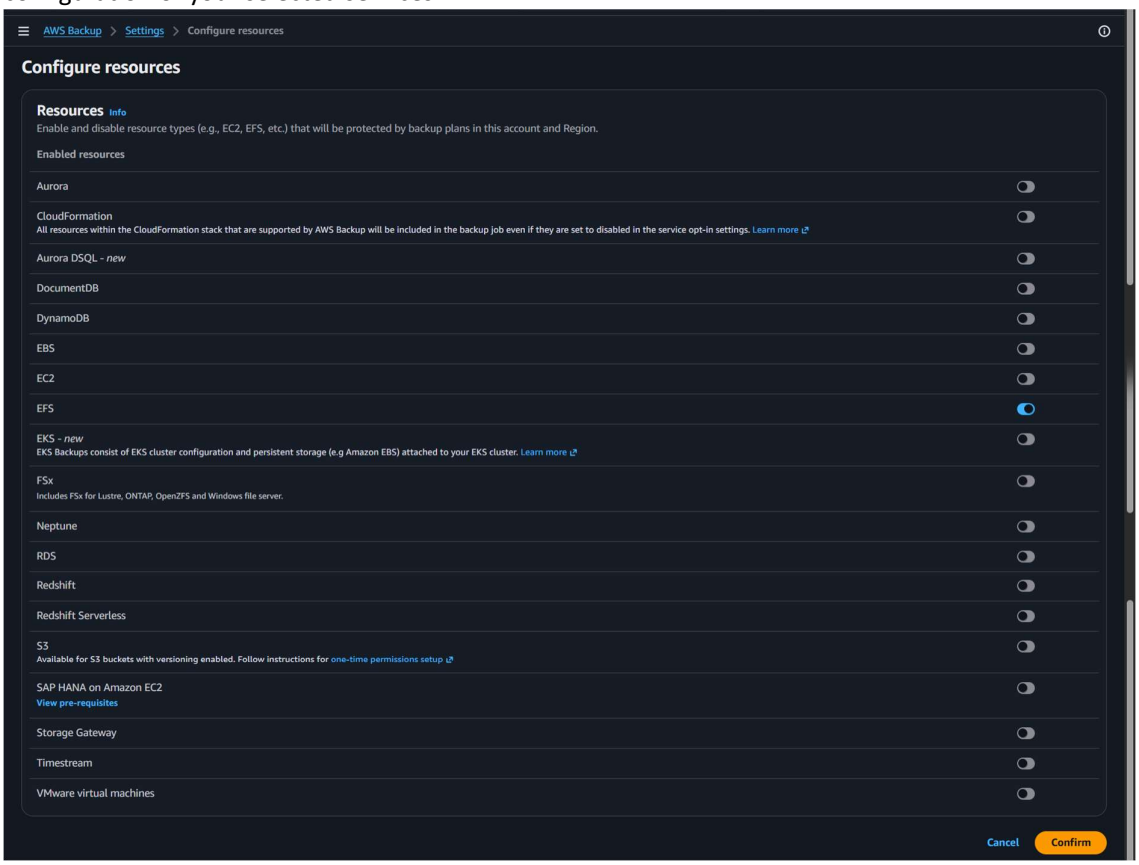
Solution

Task 1: AWS Backup On-Demand

1. In AWS management Console and open the AWS Backup
2. Set up the AWS Backup services to be utilized, choose settings and then Click on the "Configure resources" button located on the Service opt-in page.

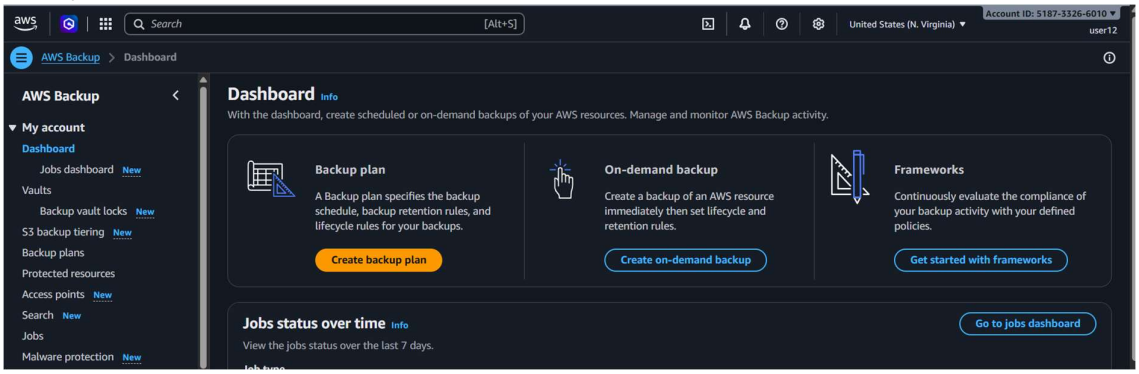


3. Access the Configure resources page and utilize the toggle switches to enable or disable the services that will be utilized with AWS Backup. Specifically, select the toggle switch for EFS. After making the necessary changes, click on the "Confirm" button to finalize the configuration of your selected services.



For this document, ensure that the AWS resources you are backing up are in the same AWS Region that you are currently using. It is important to note that the resources must be located in the same AWS Region, although there is an exception discussed in preceding steps regarding cross-Region copy. Specifically, this document will be using the US East (N. Virginia) Region (us-east-1).

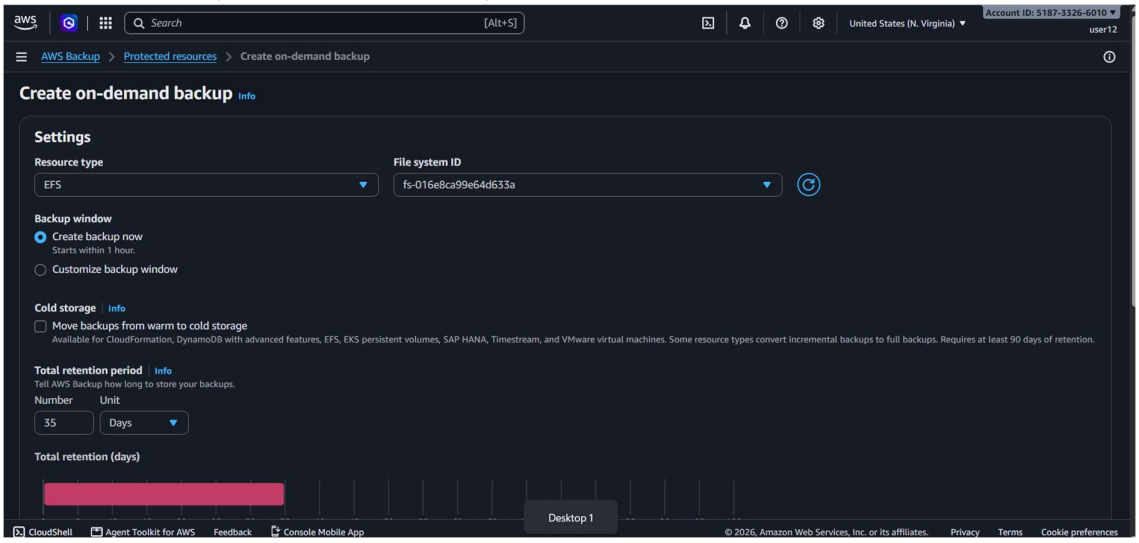
4. To proceed, navigate to the AWS Backup console and select "Dashboard" from the left-hand navigation pane located under "My account". From there, click on the "Create on-demand backup" button.

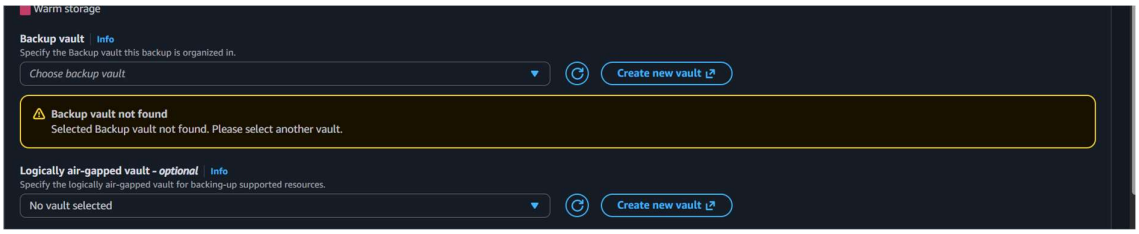


5. 5. To begin creating an on-demand backup, navigate to the "Create on-demand backup" page within the AWS Backup console there select EFS for Amazon EFS. Then, enter the File system ID for the desired EFS resource that you wish to protect.

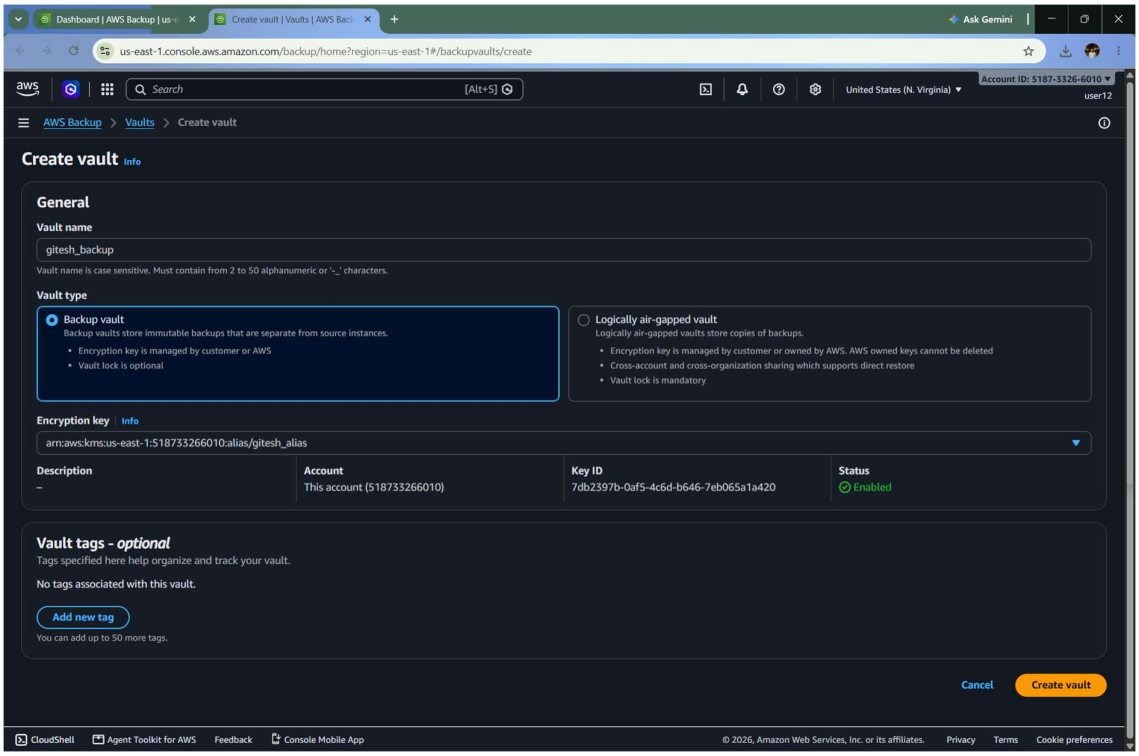
   Moving on to the Backup window section, select the "Create backup now" option to initiate an immediate backup. This allows you to access the backed-up resource sooner on the Protected resources page.

   Within the Retention period section, choose "Days" and specify the number of days you want to retain the backups for. Here chosen 7 days

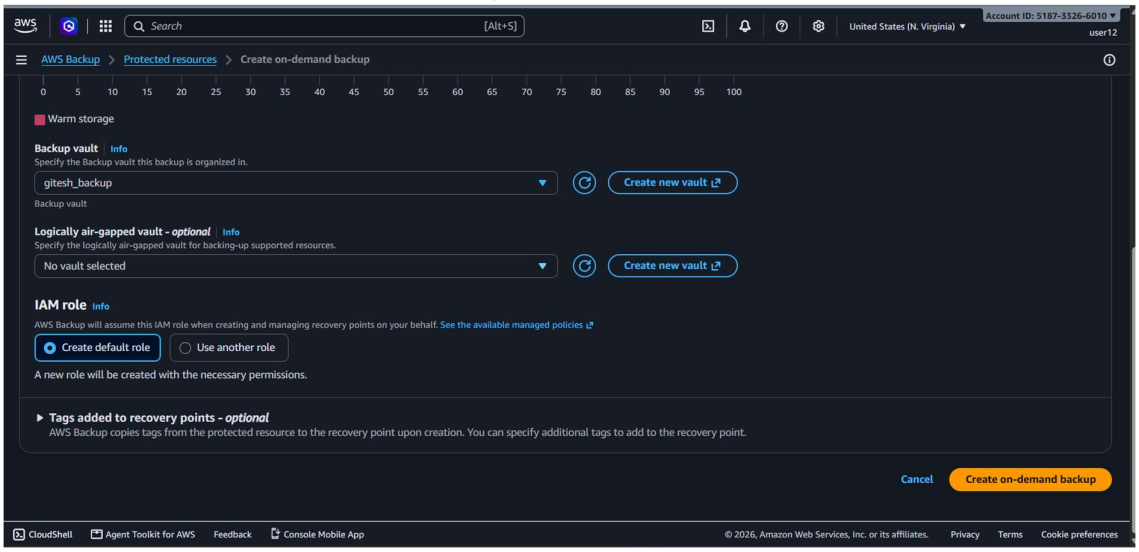




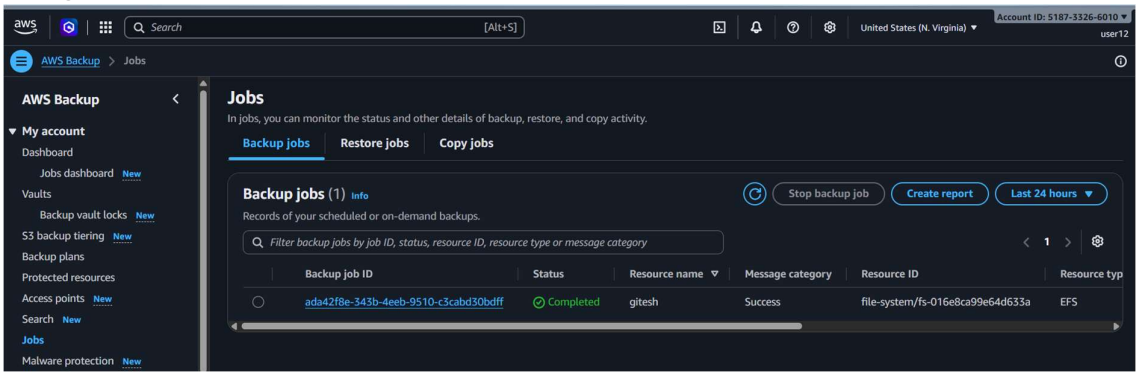
5.1 In the Backup vault section, you can either select an existing vault and continue with step 4 or create a new backup vault by selecting "Create new Backup vault" before proceeding to step 4 like shown below.



6. Now click on the Create On-Demand back-up.

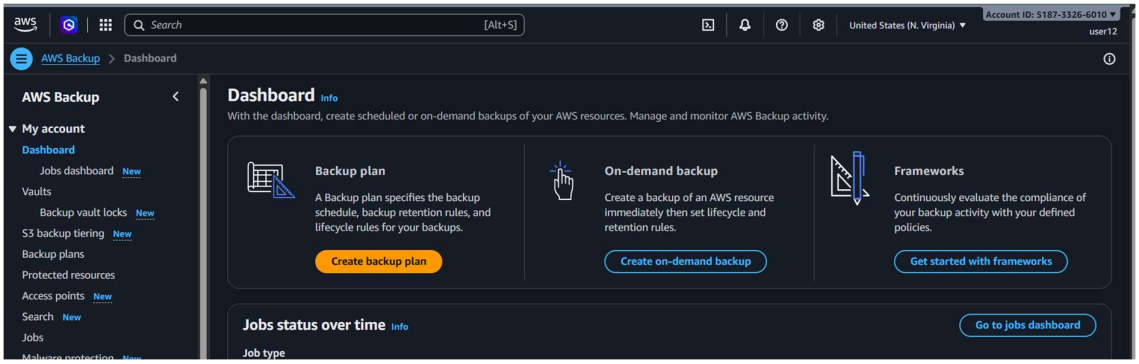


7. Now go to Jobs in the Dashboard and check the details.

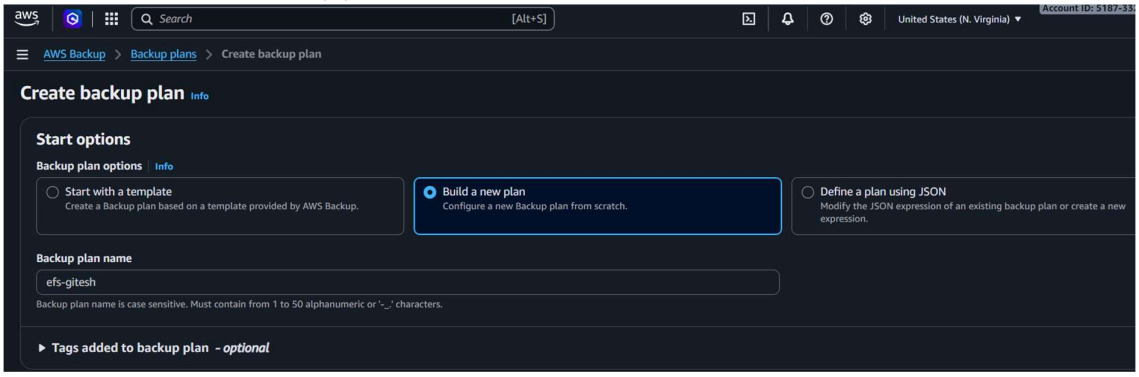


Task 2: For Configuring Automatic AWS Backup job of an Amazon EFS file systems.

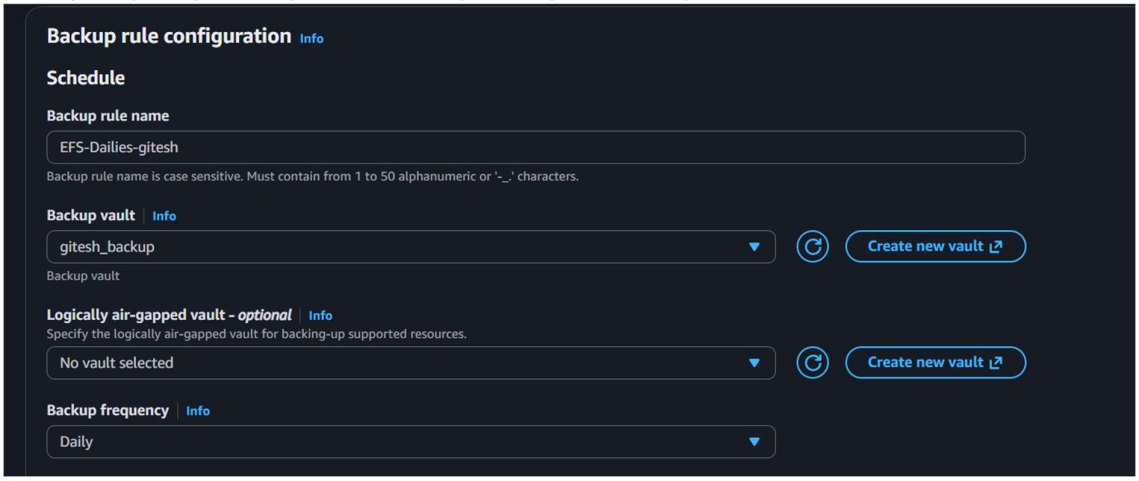
1. Repeat task 1 step 1,2,3
2. Go to Backup plans and create Backup Plans



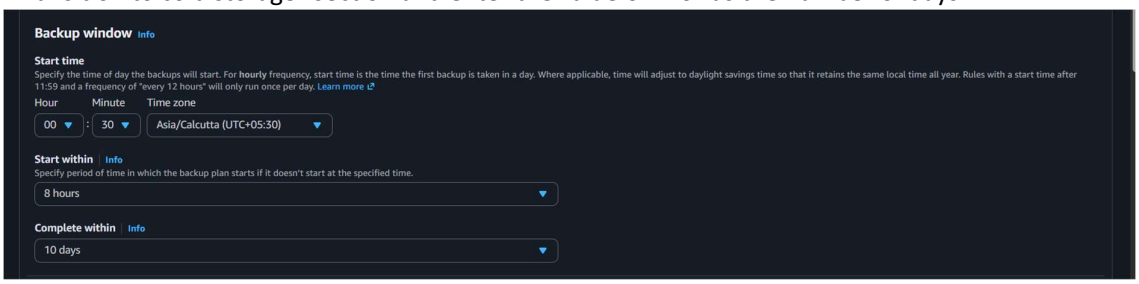
3. Give a name to the Back up plan



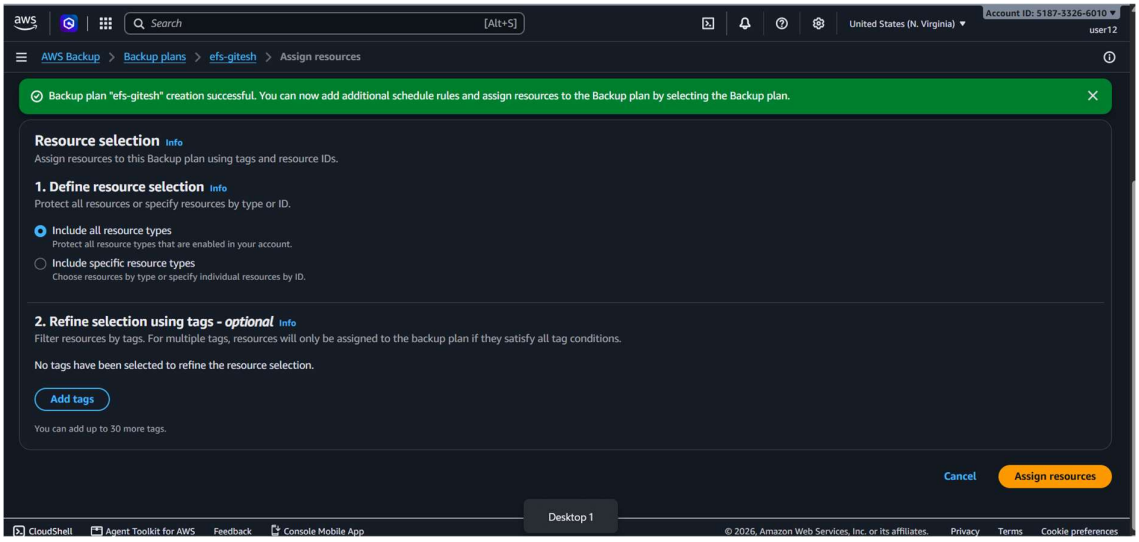
4. Create Back up Rule name then Back up vault (it can be the one already existed or can create a new one) and under the "Backup Frequency" section, select "Daily". This setting determines how frequently a backup will be generated. You have the option to choose a backup frequency of every 12 hours, daily, weekly, or monthly.



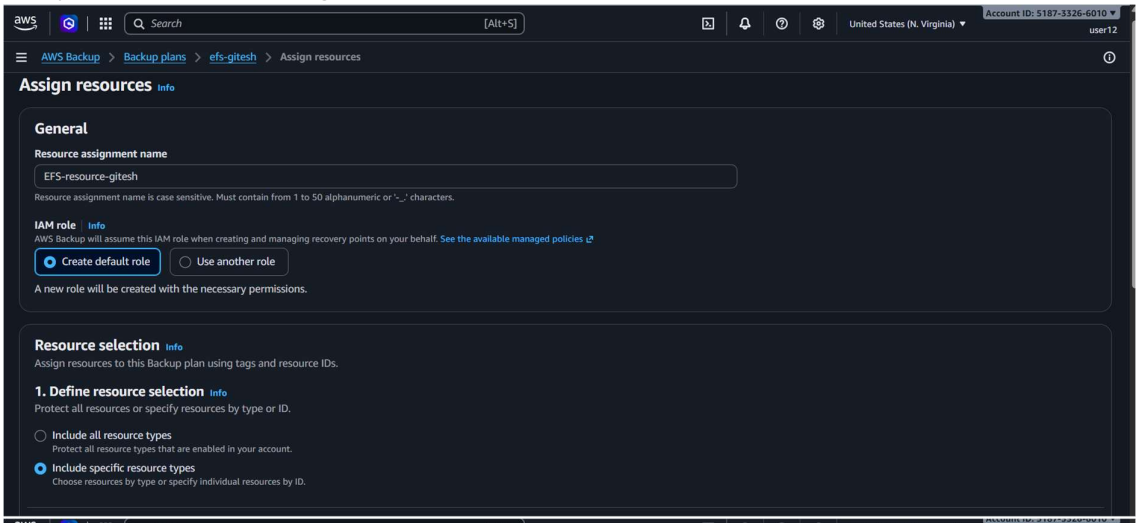
5. Within the "Backup window" section, choose the "backup window defaults" option. This will schedule the backup job to begin at 5 AM UTC (Coordinated Universal Time) and last for 8 hours.
To retain your backups in warm storage before transitioning to cold storage, go to the "Transition to cold storage" section and enter the value of "10" as the number of days.

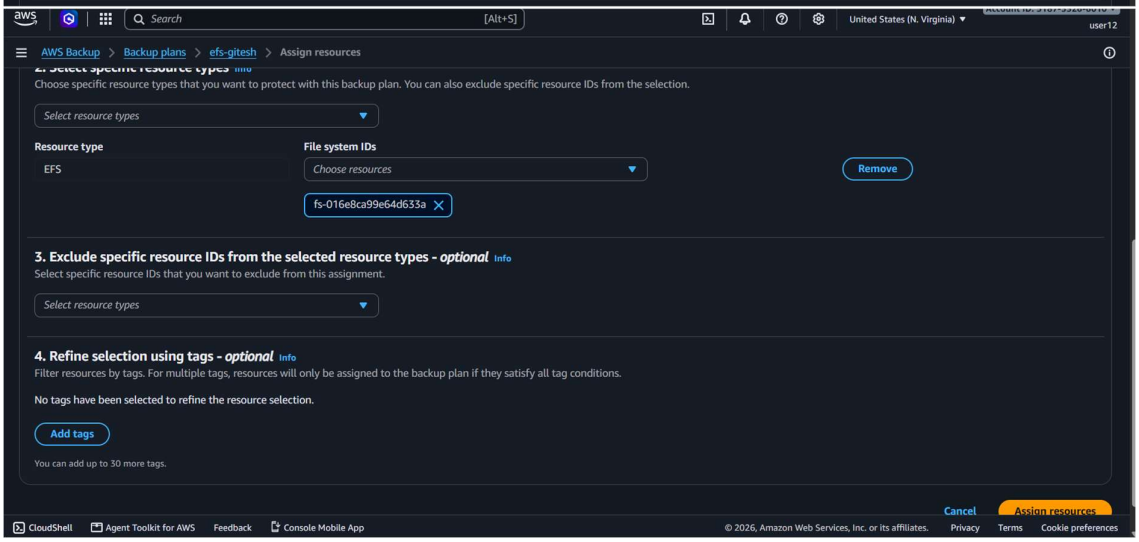


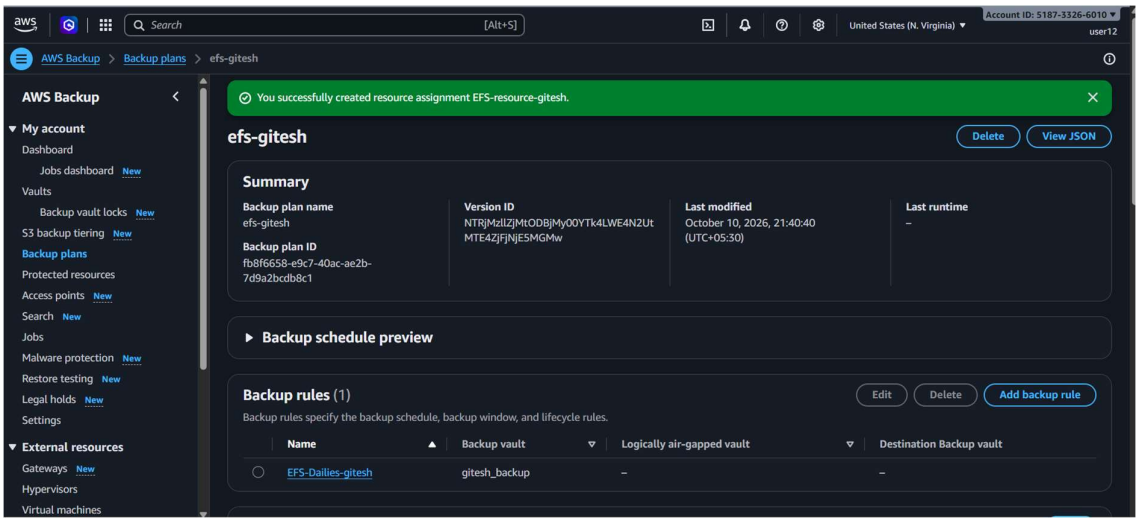
6. Now click on the create Plan.
7. Now click on Assign Resources.



8. Give Resource assignment name as “EFS-resource” and then select the specific EFS and its file system id. And click Assign Resource

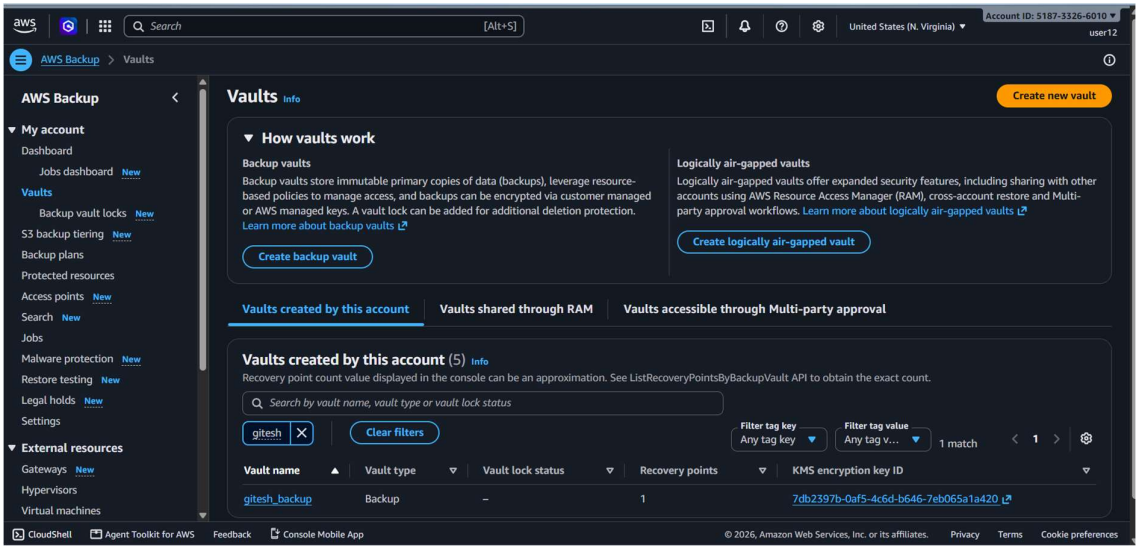




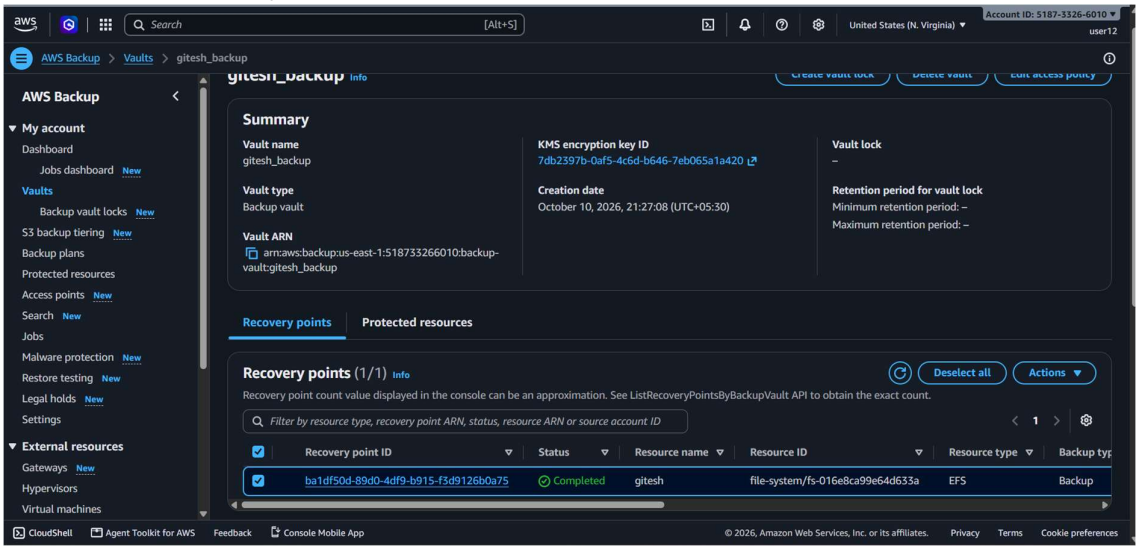


Task 3: Restoring From Backups to ensure there is no data recovery issues.

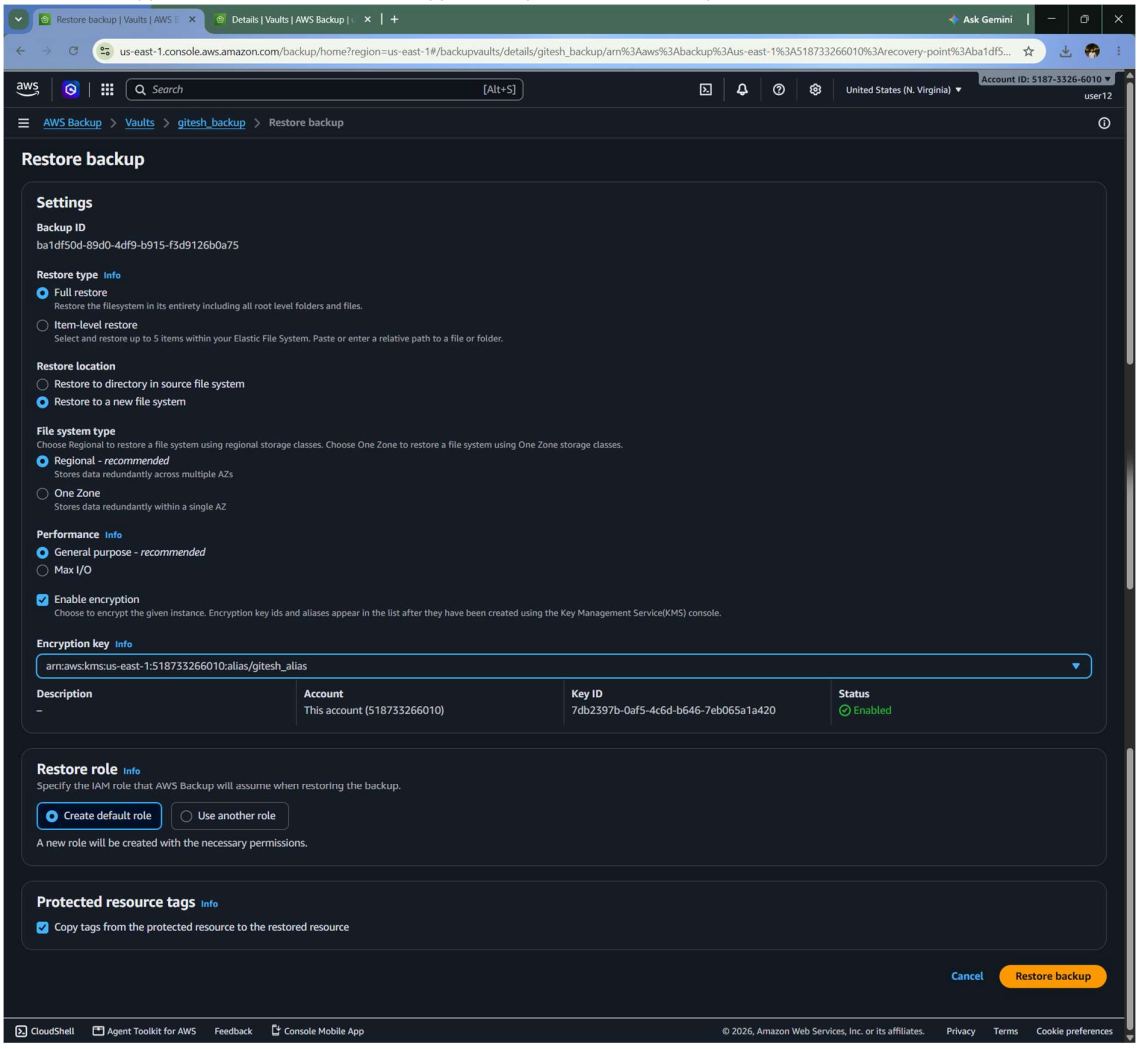
1. Go to Backup Vaults in the left panel and then select and “Webapp Backups” that we created.



2. Now select the Recovery Point Created and click on Restore.



3. Select Full restore, then Restore to a new file system, Regional, General purpose.
4. Enable Encryption and select the encrypted Key that we already created before



5. After the completion of the above step you should go back and click jobs radio button and then it should be seen like this within some time.

